

LAN Troubleshooting & Optimization

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Introduction

In today's interconnected environment, network performance directly impacts productivity and operations. Our presentation explores practical methods and tools for optimizing and troubleshooting both LAN and WAN networks. We will focus on systematic approaches to diagnose traffic issues, analyze performance bottlenecks, and implement solutions to ensure reliable connectivity. And through our practical examples, you will gain actionable insights for resolving common network problems and enhancing overall network efficiency.



Definition of LAN

Understanding Local Area Networks

A Local Area Network (LAN) connects devices within a limited geographical area, such as buildings or campuses, offering high-speed and low-latency communication compared to broader networks like WAN.

Overview of LAN Troubleshooting

Systematic Approach for Network Issues

LAN troubleshooting involves **a systematic identification, diagnosis, and repair** process to restore connectivity and performance. This structured approach ensures efficient resolution of network issues, enhancing overall productivity.

Troubleshooting Methods

Mapped to OSI Model Layers

Physical

The physical layer includes checking cables, connectors, and link lights to ensure that all hardware is functioning properly, which is critical for establishing a solid connection.

Data Link

At the data link layer, verifying MAC addresses, VLAN configurations, and Spanning Tree Protocol ensures efficient data transfer between devices while preventing loops and broadcast storms.

Network

The network layer involves checking IP configurations, routing protocols, and firewall settings to confirm that data packets are properly routed and accessible across the LAN.

The 6-Step Troubleshooting Process

The 6-step troubleshooting process provides a **structured approach** to diagnosing and resolving network issues, ensuring that technicians can effectively identify, isolate, diagnose, resolve, verify, and document problems for future reference.



Troubleshooting Tools

Command Line Tools

a. Connectivity Testing:

ping [IP] # Basic connectivity
tracert [IP] # Path discovery
pathping [IP] # Combines ping + tracert

b. Configuration Check:

ipconfig /all # Windows IP details
ifconfig # Linux IP details
arp -a # ARP table
netstat -an # Active connections
nslookup [domain] # DNS resolution

c. Switch/Router:

show interfaces # Port status (Cisco)
show ip route # Routing table
show vlan # VLAN configuration

d. Traffic Analysis Filters (Wireshark):

tcp.analysis.retransmission # Identify packet loss
tcp.analysis.duplicate_ack # Detect network congestion
icmp.type == 3 # Find "destination unreachable" errors
http.request # Analyze web traffic issues

GUI and Specialized Tools

1. **Nmap** - Network scanning
2. **PRTG** - Network monitoring
3. **SolarWinds** - Enterprise monitoring
4. **ManageEngine** - Network tools suite
5. **Advanced IP Scanner** - Quick device discovery

Common LAN Problems

Identifying and resolving frequent issues

Connectivity Faults

Connectivity faults can stem from **physical issues** like faulty cables or misconfigured devices, leading to dropped connections or inability to access the network. Regular checks are essential.

Slow Performance

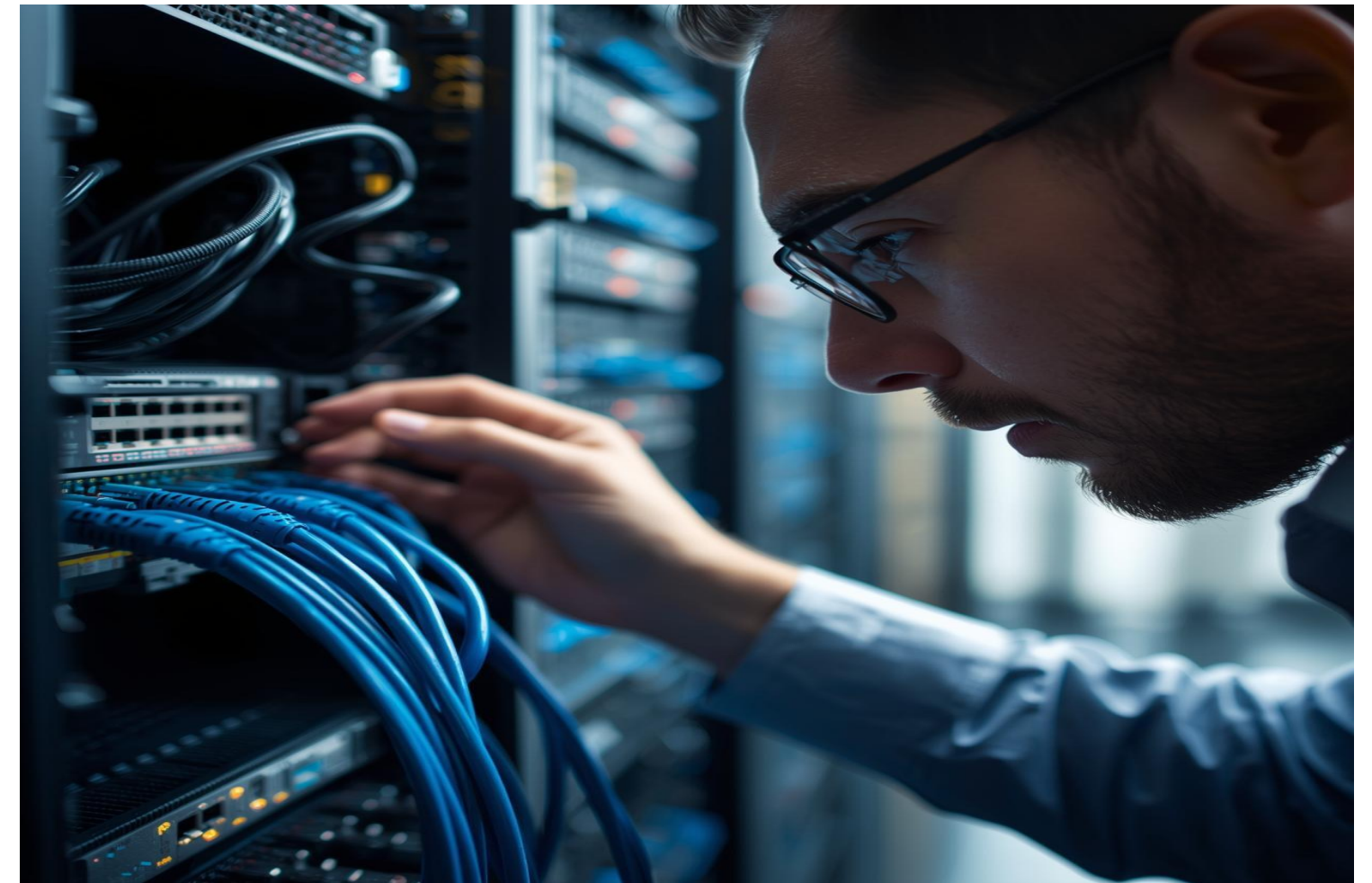
Slow performance often results from **bandwidth saturation** or duplex mismatches, causing significant delays in data transmission. Monitoring and proper configuration can mitigate these issues effectively.

Packet Loss

Packet loss occurs when data packets fail to reach their destination, leading to interruptions in communication. Common causes include network congestion, hardware faults, and faulty connections.

Troubleshooting "No Network Connectivity" with PacketTracer

Tools Used: ping, ipconfig, arp



Step 1: Test local TCP/IP stack

```
C:\>ping 127.0.0.1
```

```
Pinging 127.0.0.1 with 32 bytes of data:
```

```
Reply from 127.0.0.1: bytes=32 time=3ms TTL=128  
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128  
Reply from 127.0.0.1: bytes=32 time=3ms TTL=128  
Reply from 127.0.0.1: bytes=32 time=3ms TTL=128
```

```
Ping statistics for 127.0.0.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 3ms, Average = 2ms
```

Success = TCP/IP working locally

Step 2: Test own IP

```
C:\>ping 192.168.1.103
```

```
Pinging 192.168.1.103 with 32 bytes of data:
```

```
Reply from 192.168.1.103: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.103: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.103: bytes=32 time=2ms TTL=128  
Reply from 192.168.1.103: bytes=32 time=3ms TTL=128
```

```
Ping statistics for 192.168.1.103:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Success = NIC configured

- **Step 3: Test gateway**

```
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Failure = Problem with LAN (IDENTIFIED)

- **Step 4: Check IP configuration**

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:8FFF:FE2D:7892
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.103
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.0.1
```

Look For: IP Address, Subnet, Gateway

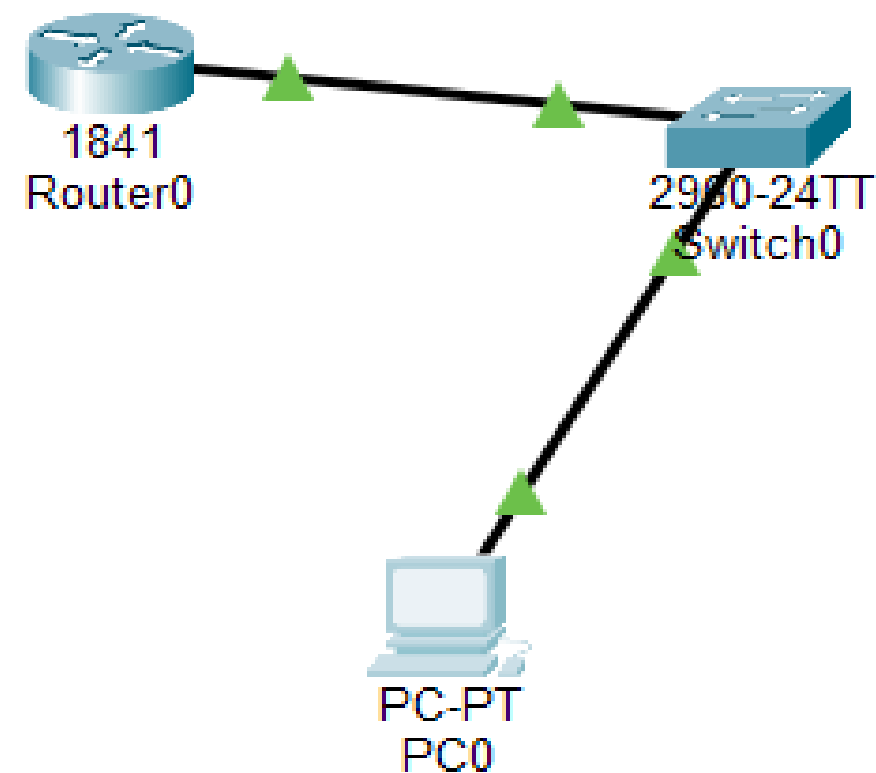
ISOLATED THE PROBLEM! IP Is 192.168.1.103, Gateway Is 192.168.0.1 - Different Subnets

- **Step 5: Check ARP table**

```
C:\>arp -a  
No ARP Entries Found  
C:\>
```

ISOLATED another problem (doesn't show gateway MAC address) as PC can't ARP for gateway (different subnet)

- **Step 6: Physical check**



GREEN LIGHTS -no cable issues visible

- **DIAGNOSING Causes & Fixes:**

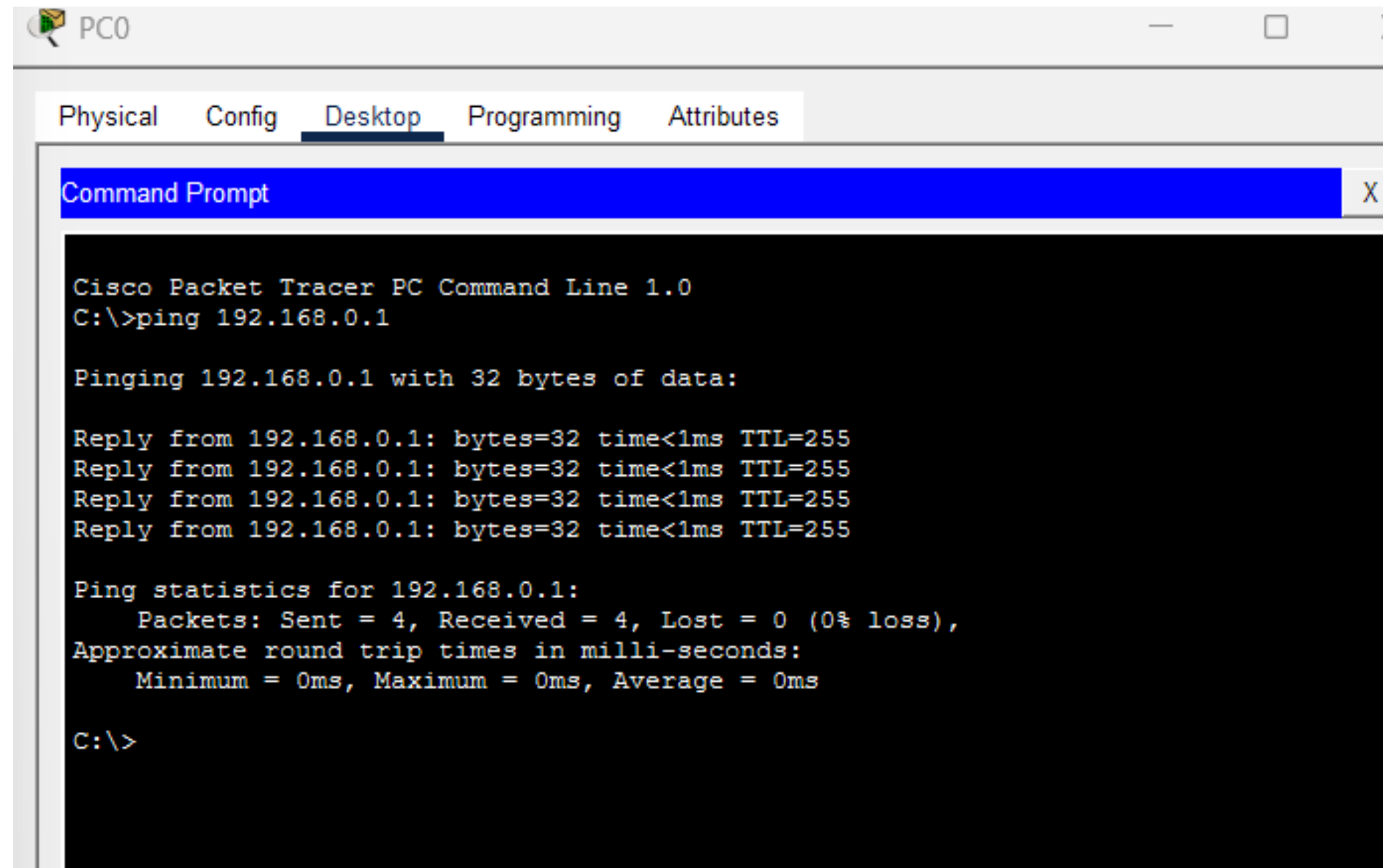
Cause – Layer 3 problem- IP misconfiguration.

- **Symptoms:** Intermittent connectivity

- **Fix: Change IP or enable DHCP snooping**

IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.0.103
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
DNS Server	0.0.0.0

- **Test 1: Ping Gateway**



The screenshot shows a PC window titled 'PC0' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Result: Reply from 192.168.0.1... → Success!

- **Test 2: Check ARP Table**

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.0.1           0030.a370.dc01        dynamic
```

Result: Shows Gateway MAC Address → ARP Working!

Problem solved!

LAN Optimization :

Improving Network Performance and Efficiency.

The primary goals of LAN optimization include reducing **latency**, increasing throughput, enhancing reliability, improving security, and simplifying management to ensure a more efficient network environment.

Optimization Methods

Key STRATEGIES for improving LAN performance

Traffic Management

Quality of Service (QoS) techniques help prioritize network traffic, ensuring critical applications receive necessary bandwidth, which reduces latency and enhances overall user experience across the LAN.

Network Segmentation

Implementing VLANs allows for logical separation of network traffic, reducing broadcast domains and improving performance by isolating different departments or functions within the same physical infrastructure.

Infrastructure Tuning

Implementing STP optimization ensures faster network recovery, prevents loops, and optimizes traffic flow by strategically configuring protocol settings, root bridges, and port behaviors within the existing physical infrastructure.

Optimization Tools

Monitoring & Analysis Tools:

1. SolarWinds NPM - Network Performance Monitor
2. PRTG Network Monitor - All-in-one monitoring
3. Wireshark - Protocol analysis
4. NetFlow Analyzer - Traffic analysis
5. Nagios - Open-source monitoring

Configuration & Management Tools:

1. Cisco DNA Center - Intent-based networking
2. SolarWinds NCM - Configuration management
3. ManageEngine - Network tools suite
4. Ansible/Puppet - Automation

Bandwidth Testing Tools:

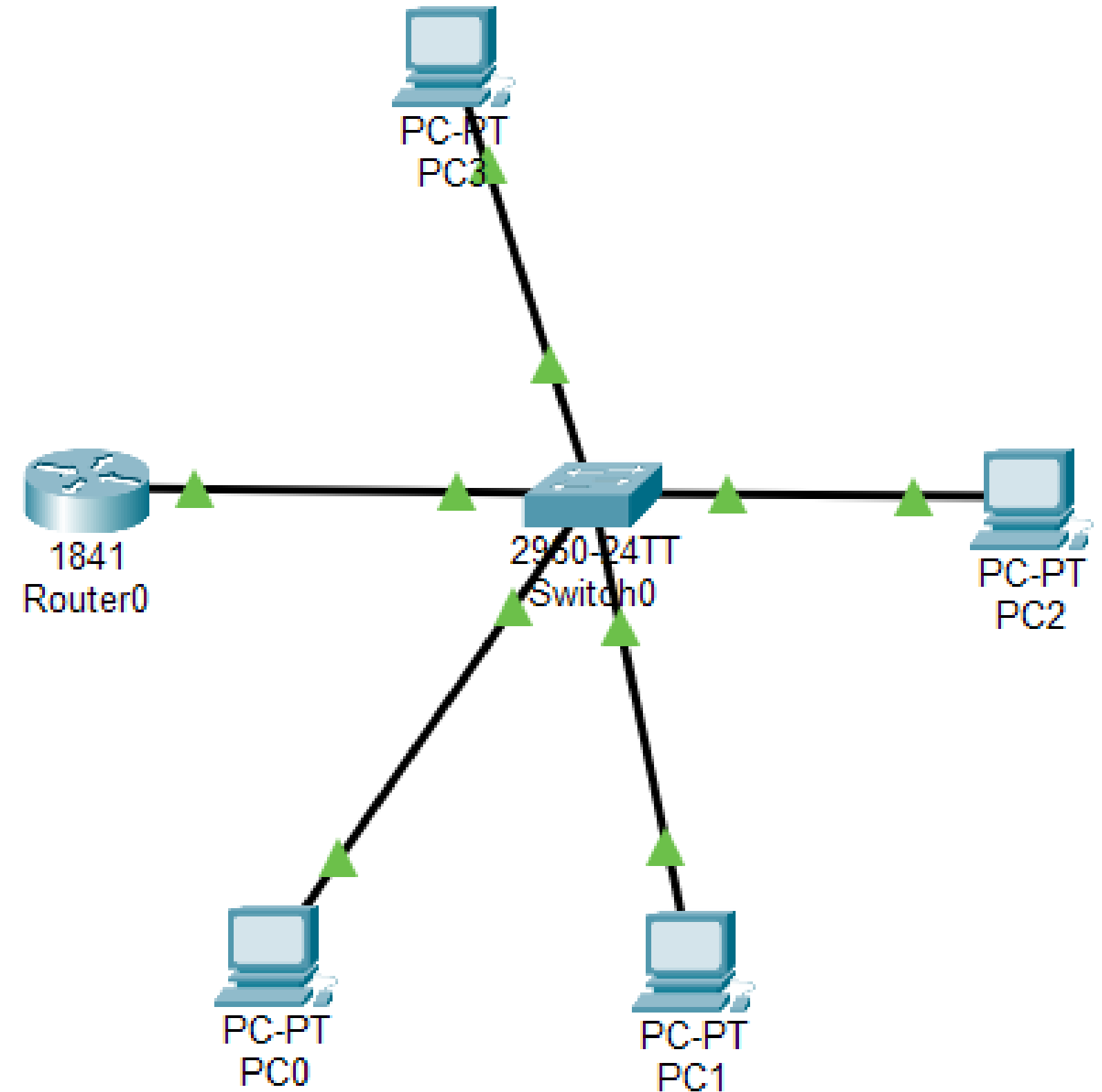
1. iperf3 -s - Server mode
2. iperf3 -c server-ip - Client mode

VLAN Implementation

Example for “FLAT NETWORK” with PacketTracer

- **Initial State:**

- ❑ 6 devices (1 Router, 1 Switch, 4 PC)



STEP-1: CHECKING BROADCAST

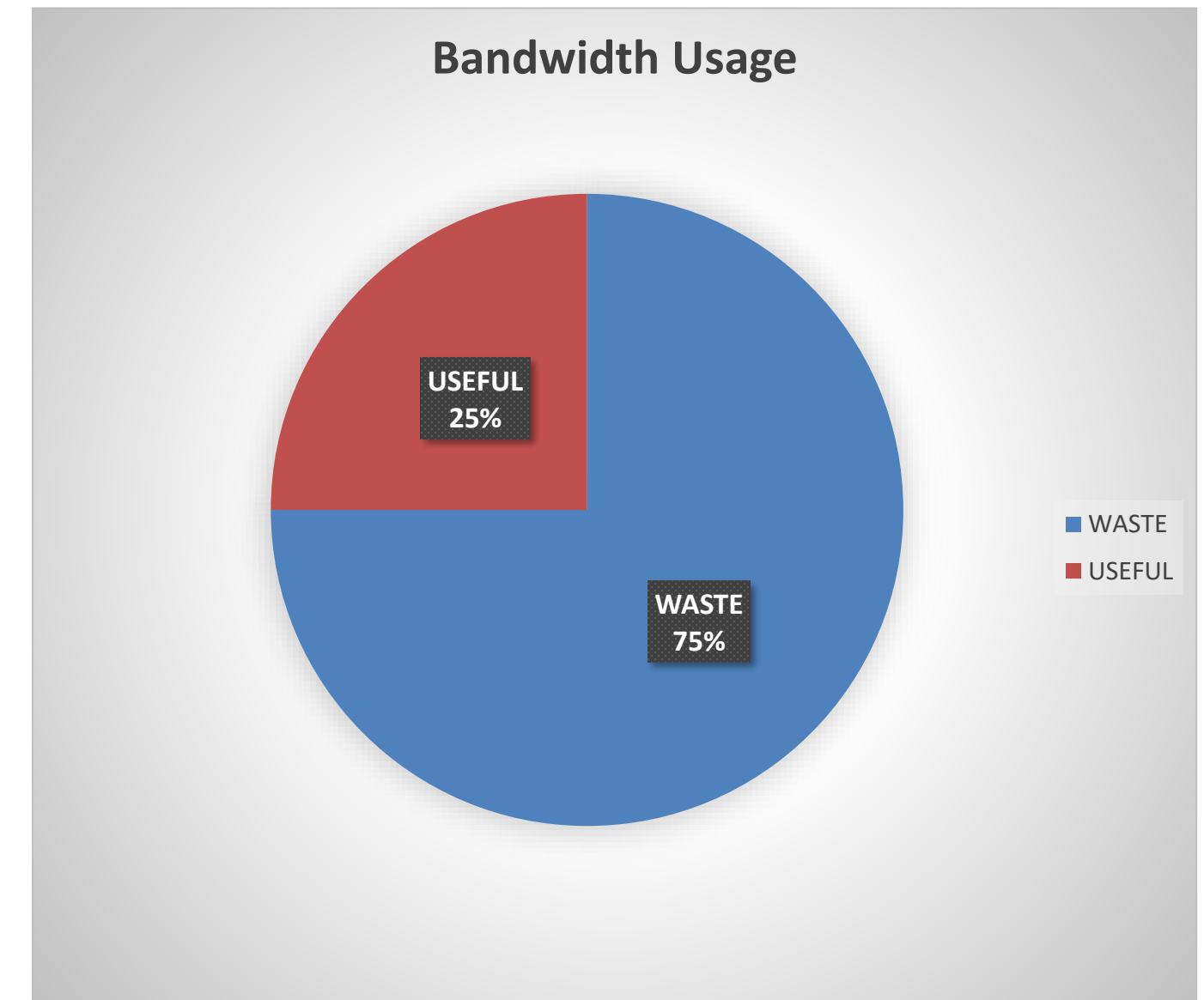
- Demonstrating our current network problem. When any device(here PC0) sends broadcast traffic..

```
C:\>ping 192.168.1.255 -n 5

Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 5, Received = 20, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



Problem: Broadcast reaches all devices

Evidence: Our test shows 4/4 devices receive

Impact: Wastes 75% bandwidth

OPTIMIZATION Procedure :

Step 1: Implement VLANs

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname CoreSwitch
CoreSwitch(config)#vlan 10
CoreSwitch(config-vlan)#name SALES
CoreSwitch(config-vlan)#exit
CoreSwitch(config)#vlan 20
CoreSwitch(config-vlan)#name ENGINEERING
CoreSwitch(config-vlan)#exit
CoreSwitch(config)#interface range FastEthernet0/1-2
CoreSwitch(config-if-range)#switchport mode access
CoreSwitch(config-if-range)#switchport access vlan 10
CoreSwitch(config-if-range)#exit
CoreSwitch(config)#interface range FastEthernet0/3-4
CoreSwitch(config-if-range)#switchport mode access
CoreSwitch(config-if-range)#switchport access vlan 20
CoreSwitch(config-if-range)#exit
CoreSwitch(config)#interface FastEthernet0/24
CoreSwitch(config-if)#switchport mode trunk

CoreSwitch(config-if)#switchport trunk allowed vlan 10,20
CoreSwitch(config-if)#exit
CoreSwitch(config)#end
CoreSwitch#write memory
```

Step 2: Router configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Gateway
Gateway(config)#interface FastEthernet0/0
Gateway(config-if)#no ip address
Gateway(config-if)#exit
Gateway(config)#interface FastEthernet0/0.10
Gateway(config-subif)#encapsulation dot1Q 10
Gateway(config-subif)#ip address 192.168.10.1 255.255.255.0
Gateway(config-subif)#exit
Gateway(config)#interface FastEthernet0/0.20
Gateway(config-subif)#encapsulation dot1Q 20
Gateway(config-subif)#ip address 192.168.20.1 255.255.255.0
Gateway(config-subif)#exit
Gateway(config)#interface FastEthernet0/0
Gateway(config-if)#no shutdown
Gateway(config-if)#exit
Gateway(config)#end
Gateway#write memory
```

Step 3: Updating PC IPs:

PC0 - PC1 (VLAN 10):

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

PC2 – PC3 (VLAN 20):

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.12

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.13

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

Results After Optimization:

On PC0 (VLAN 10):

```
C:\>ping 192.168.10.255 -n 5

Pinging 192.168.10.255 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.255:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.255 -n 5

Pinging 192.168.20.255 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.255:
    Packets: Sent = 5, Received = 0, Lost = 5 (100% loss),
```

Only PC0 & PC1 receive

On PC2 (VLAN 20):

```
C:\>ping 192.168.20.255 -n 5

Pinging 192.168.20.255 with 32 bytes of data:

Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time=6ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.255:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ping 192.168.10.255 -n 5

Pinging 192.168.10.255 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.255:
    Packets: Sent = 5, Received = 0, Lost = 5 (100% loss),
```

Only PC2 & PC3 receive

Performance Analysis Results:

1. Broadcast Domain Reduction:

- Before: 1 broadcast domain with 4 devices
- After: 2 broadcast domains with 2 devices each
- Result: 50% reduction in broadcast traffic per domain

2. Traffic Isolation:

- Before: All devices receive all broadcasts
- After: VLAN 10 broadcasts only affect Sales department and VLAN 20 broadcasts only affect Engineering department
- Result: Through departments separation , they can't accidentally receive each other's traffic.

Best Practices

Effective Strategies for LAN Performance

Broadcast Storms

Broadcast storms can severely impact network performance, resulting from excessive broadcast traffic. Implementing VLANs and optimizing Spanning Tree Protocol (STP) can effectively mitigate these issues.

Duplex Mismatches

A common problem in LANs, **duplex mismatches** occur when devices operate at different duplex modes. Forcing devices to full-duplex can significantly enhance network performance and avoid collisions.

ARP Cache Poisoning

ARP cache poisoning is a security threat where attackers send fake ARP messages. Employing DHCP snooping and ARP inspection can help protect against unauthorized access and ensure network integrity.

The 24-Hour Test

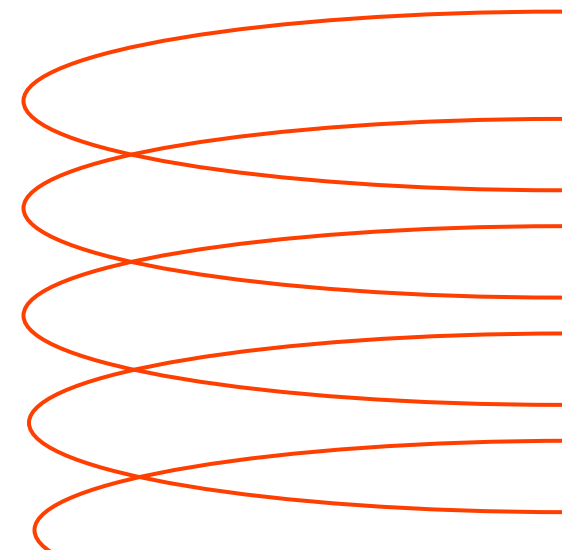
Measure Improvements After Optimization

Conducting the **24-hour test** involves assessing network performance improvements within a day of making optimizations. If no enhancements are observed, it's essential to revert and reassess.

Conclusion on LAN and WAN Troubleshooting and Optimization

In conclusion, effective troubleshooting and optimization of **LAN** are critical to maintaining **network efficiency** and **performance**. Organizations must prioritize the following:

- Regular monitoring of network performance.
- Identifying potential bottlenecks.
- Analyzing traffic patterns.
- Implementing proactive measures for device maintenance.
- Regular updates to firmware and software.
- Utilizing the right tools for diagnostics and analysis.



References:

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